Tumor-related Lipid Exudation after Plaque Radiotherapy of Choroidal Melanoma: The Role of Bruch's Membrane Rupture

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Purpose: To identify the risk factors predictive of development of tumor-related lipid exudation (TRLE) after plaque radiotherapy of posterior uveal melanoma.

Design: Case-control study.

Participants: Cases included 294 patients with posterior uveal melanoma who had developed TRLE after plaque radiotherapy. Controls included 294 patients with posterior uveal melanoma who had not developed TRLE after plaque radiotherapy. Controls were matched with cases for age, gender, and initial tumor thickness.

Methods: Data were extracted from medical charts containing demographic, clinical, and treatment information. Detailed fundus drawings and color fundus photographs were reviewed for each patient.

Main Outcome Measures: Tumor and ocular features of eyes with posterior uveal melanoma treated with plaque radiotherapy.

Results: Multivariate analysis identified Bruch's membrane rupture ($P<0.001$), serous retinal detachment (RD) before radiation ($P\leq0.019$), closer proximity to the optic disc and foveola ($P = 0.004$ and 0.013, respectively), greater tumor base ($P = 0.035$), failure to receive transpupillary thermotherapy (TTT) after radiation ($P<0.001$), and initial increase of serous RD after radiation ($P<0.001$) as significant risk factors predictive of development of TRLE after plaque radiotherapy of posterior uveal melanoma. Radiation dose at the tumor base correlated with maximum extent of TRLE ($P = 0.003$). The mean interval between plaque radiotherapy and onset of TRLE was 14 months (median, 11 months; range, 2–97 months), with 88% of cases developing TRLE within 2 years of radiation. The interval between the onset of TRLE and the first evidence of its regression was a mean of 33 months (median, 38 months; range, 2–194 months).

Conclusions: Our study identified Bruch's membrane rupture as an important factor predisposing to development of TRLE after plaque radiotherapy of posterior uveal melanoma. Other predictive factors included serous RD before radiation, large tumor basal diameter, posterior tumor location, lack of adjunctive TTT, and early increase of serous RD after plaque radiotherapy.

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Radiation, generally in the form of brachytherapy,[1] proton-beam,[2,3] or gamma knife,[4] is a common method for treatment of small and medium-sized posterior uveal melanomas and is a viable option for treatment of larger melanomas in selected cases.[5–8]

Despite a high rate of local tumor control,[1–8] all forms of radiotherapy can lead to complications such as glaucoma, cataract, vitreous hemorrhage, retinopathy/maculopathy, and papillopathy.[9–15] The clinical features[11] and the systemic and ocular conditions predisposing to development of radiation-induced retinopathy/maculopathy and papillopathy[10,12] have been reported. A less-known potential complication of radiotherapy of posterior uveal melanoma is exudation of lipid from the irradiated residual tumor,[16] a condition that we refer to as "tumor-related lipid exudation" (TRLE). This case-control study was designed to identify the systemic and tumor features predictive of development of TRLE after plaque radiotherapy of posterior uveal melanoma.

Materials and Methods

All cases of posterior uveal melanoma that were managed at the Oncology Service, Wills Eye Institute, between 1978 and 2007, and presented with or developed TRLE after plaque radiotherapy were included in this study. Eyes with serous retinal detachment (RD) without visible lipid deposition and eyes with lipid exudation that, on the basis of clinical or fluorescein angiographic findings, originated from abnormal retinal or optic disc vessels (radiation-induced retinopathy/maculopathy or papillopathy, respectively), and not from the treated melanoma, were excluded from the study.

Controls were chosen from patients with posterior uveal melanoma managed with plaque radiotherapy who had not developed TRLE after radiation. Controls were matched with cases for age (within 5 years), gender, and tumor thickness (within 1 mm).

Our methods of clinical examination, diagnosis, and treatment have been described.[9] Plaque radiotherapy was performed using iodine-125 or, in some earlier cases, cobalt-60 delivering a planned dose of 8000 centigray (cGy) to the tumor apex. Adjunct transpupillary thermotherapy (TTT) was used for many

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